

Idea Description

A ORB - Omni Reconnaissance Bot

Our Idea described the designing of an a robotic bot which will be equipped with Multiple sensors and Imaging technologies.

For the identification of humans and various object detection in various climate conditions.

A compact palm-sized spherical device equipped with visible light cameras, thermal infrared sensors, and laser range finders for 360-degree surveillance.

A smartphone-sized surveillance tool disguised as a regular mobile device, integrating advanced multi-spectral sensors and wireless communication capabilities

The Spherical orb will be lightweight and attachable to a drone, Stick, capable of detecting human-sized targets up to 1.5 km by day and 1 km by night, even in adverse weather.

The ORB (Omni Reconnaissance Bot) is a cutting-edge surveillance solution aimed at providing robust, reliable monitoring in extreme weather and challenging operational environments.

Created for applications such as border security, tactical reconnaissance, and disaster management, the ORB system consists of a compact, palm-sized spherical device and a discrete smartphone-sized variant.

These devices are engineered to deliver omni-directional, 360-degree situational awareness using a fusion of visible light cameras, long-wave infrared (LWIR) thermal sensors, and precision laser range finders—all embedded within a durable metallic casing optimized for field conditions.

At the heart of the ORB’s technology is sensor integration and fusion.

The system’s multiple imaging and ranging modules continuously collect environmental data, which is synchronized and processed via embedded signal conditioning and sensor fusion algorithms.

This enables real-time, multi-sensor data streams—such as high-resolution visual feeds, thermal imagery for heat signature detection, and accurate distance measurements from the laser range finder—to be blended into a single, comprehensive operational picture.

Power is supplied by lightweight, rechargeable lithium-polymer batteries crafted for extended missions, particularly when the device is drone-mounted.

The ORB is designed with modularity in mind: it can be handheld, fixed to a pole, or attached to unmanned aerial vehicles, allowing deployment across diverse terrains and scenarios.

Data transmission is another cornerstone of the ORB’s design. Leveraging mature and widely available wireless communication modules, the device can stream compressed, encoded data in real-time to ground stations or operators via Wi-Fi, LTE, or 5G.

This data can then be viewed on mobile devices, augmented reality HUDs, or centralized surveillance systems.

The communication backbone is engineered for low latency, real-time responsiveness, and adaptability with software-defined radios, ensuring the ORB remains effective even in environments with variable connectivity.

User interface considerations include intuitive mobile and AR-based controls, enabling operators to pan the device, switch sensor modes, and access intelligence overlays seamlessly.

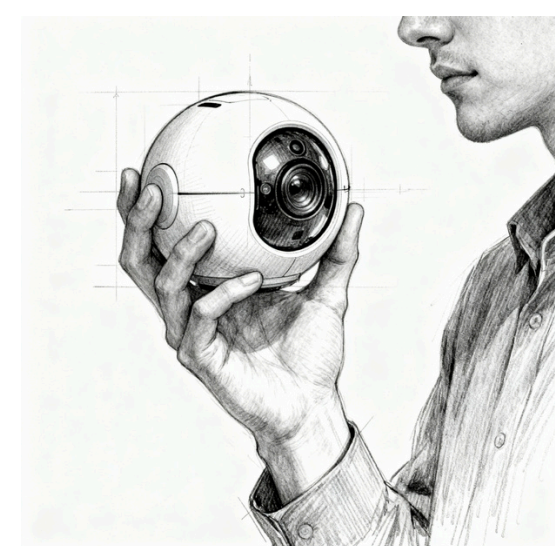
The onboard AI processor assists in rapid target detection and classification—reducing operator workload and sharpening mission focus, especially in high-stress tactical environments.

Technical feasibility is underscored by the availability of market-ready sensor modules (thermal, visible, LRF), compact optics, advanced AI processors, and robust communication tech, minimizing R&D effort and accelerating deployment.

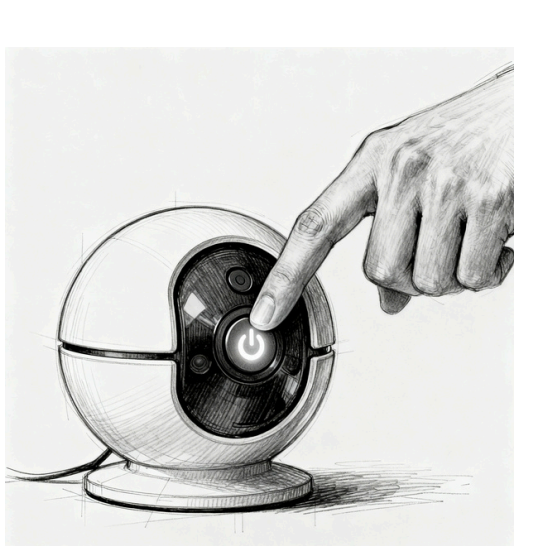
Risks such as sensor calibration or environmental noise are addressed through robust mechanical and signal-processing design, iterative field testing, and comprehensive user training programs.

Using Principle of the Surveillance Device

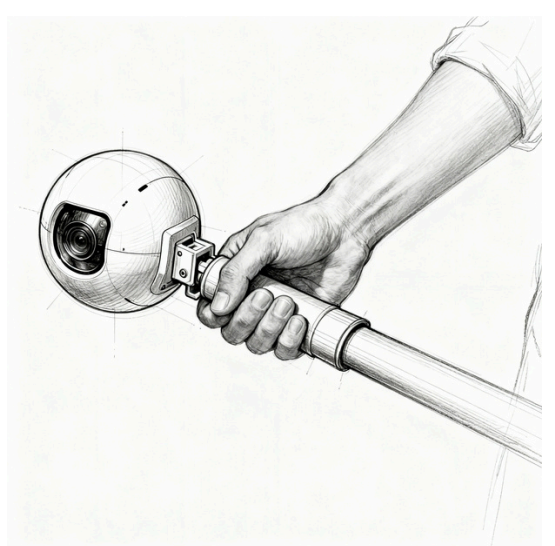
How to Operate the Surveillance Device



Holding the Surveillance Device



Switch ON the Device



Attachment to pole,(Drone in other cases)



Raising the Surveillance Device



Connecting to the Surveillance Device



Establing Communication channel



Live Feed on User Display



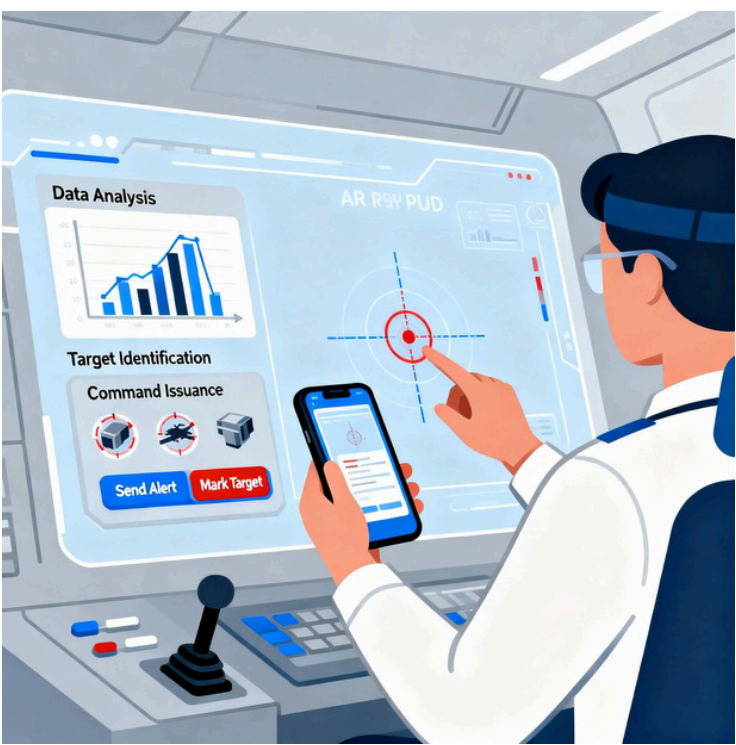
View From the Surveillance Device



After Work,Bringing Down the Device



Removing the Onboard Memory



Post Analysis from the Memory

Operation of the Spherical Omni-Directional Surveillance Orb

Power On: The user takes the small spherical orb in hand and switches it on by pressing the power button or activating it via a connected mobile app or remote control.

Mount or Hold: The orb can be handheld for close-range surveillance or attached securely to an extendable pole or drone mounting bracket for aerial or high-angle views.

Connect to Ground Station: Using wireless communication (Wi-Fi, GSM, LoRa, etc.), the orb transmits live data to a mobile device, tablet, or dedicated ground station for monitoring.

View Feed and Controls: The operator views live video feeds and sensor data on a mobile app, augmented reality HUD, or monitor, using an intuitive interface for pan/tilt control, zooming, and sensor mode switching (visible, thermal, laser range).

Target Detection: AI-driven analytics highlight detected persons, vehicles, or other objects of interest automatically on the feed.

Record or Stream: The user can record data locally on the device or stream it to cloud storage or command centers.

Power Off & Secure: After use, the device is powered down and safely stored or redeployed.

The devices are designed for intuitive operation with minimal training.

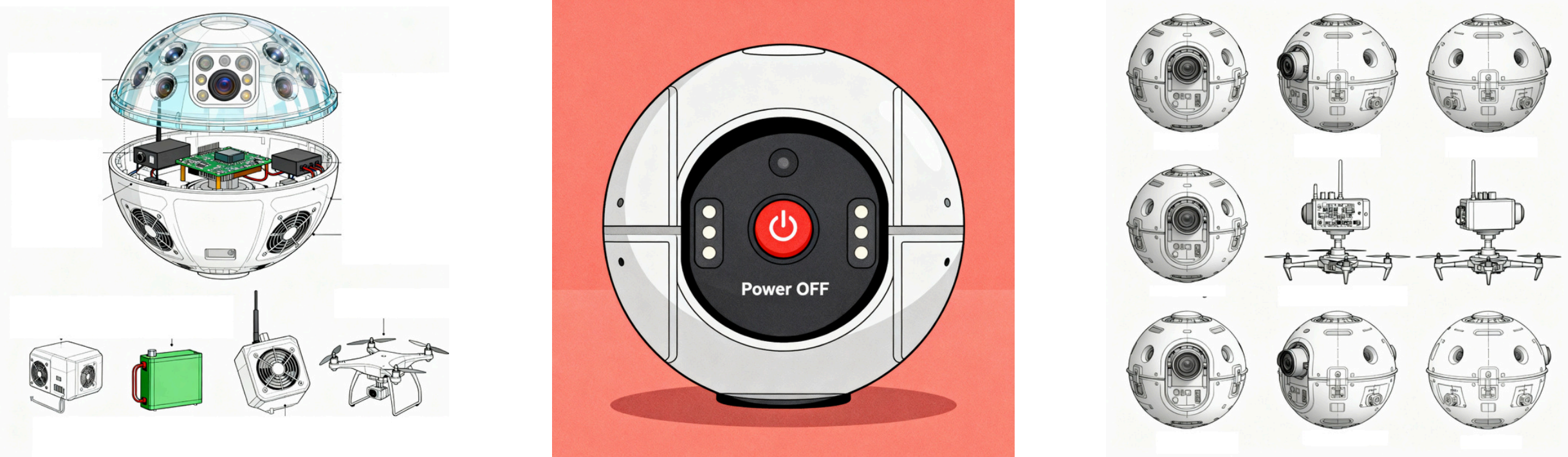
Users interact primarily through mobile or AR-based apps that provide live multi-sensor data, AI-assisted detection, and remote control.

The small form factor and wireless design facilitate rapid deployment in diverse environments, whether handheld, pole-mounted, or drone-carried.

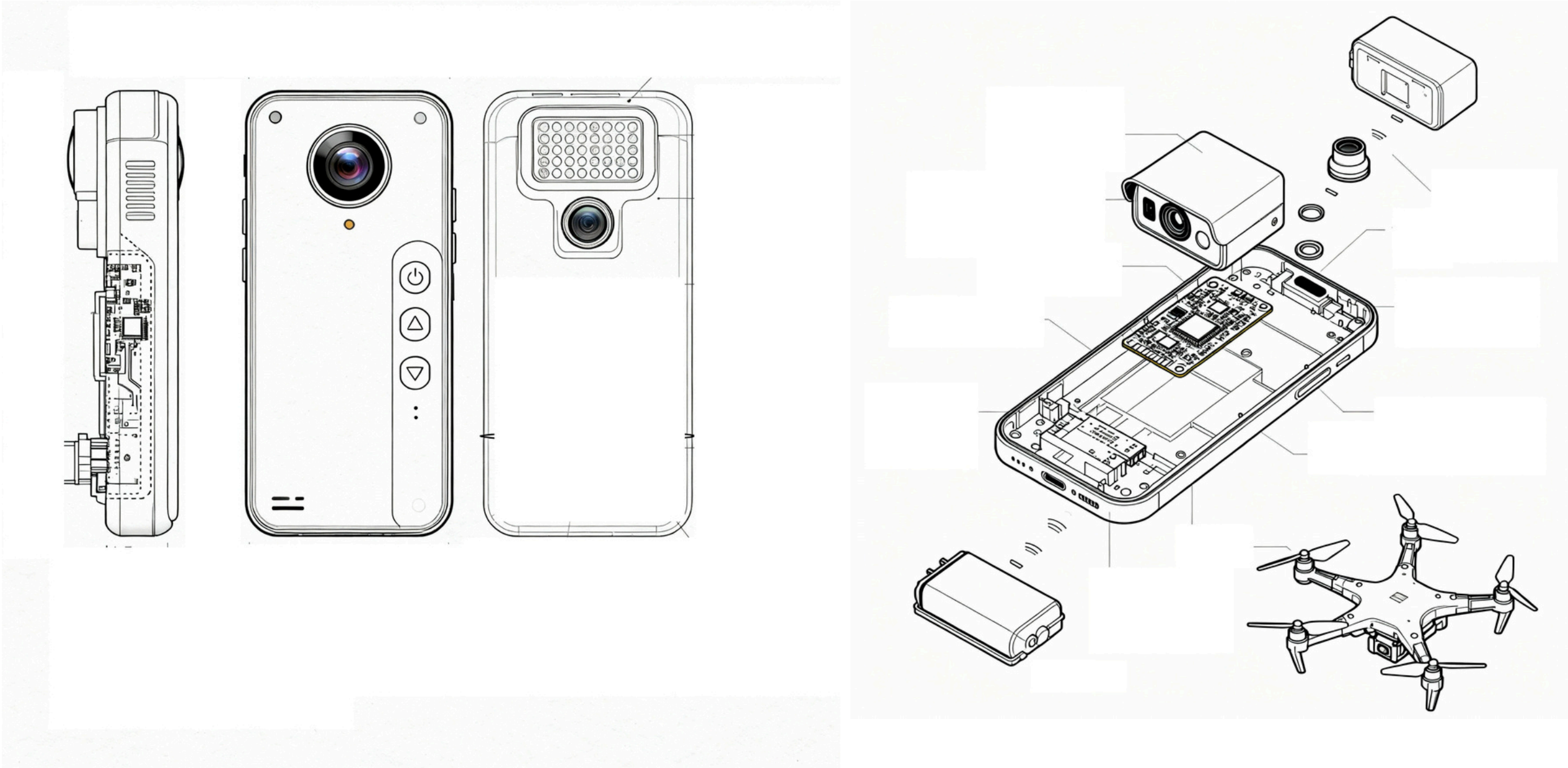
A person operates these surveillance devices by powering them on via a button or connected app, then either holding the spherical orb or attaching it to a pole or drone.

Using a mobile app, tablet, or AR HUD, the user views live multi-sensor feeds (visible, thermal, laser) and controls sensor modes remotely.

Conceptual Design For the Surveillance Device-Sphere(Orb) Design



Conceptual Design For the Surveillance Device-Mobile style Design



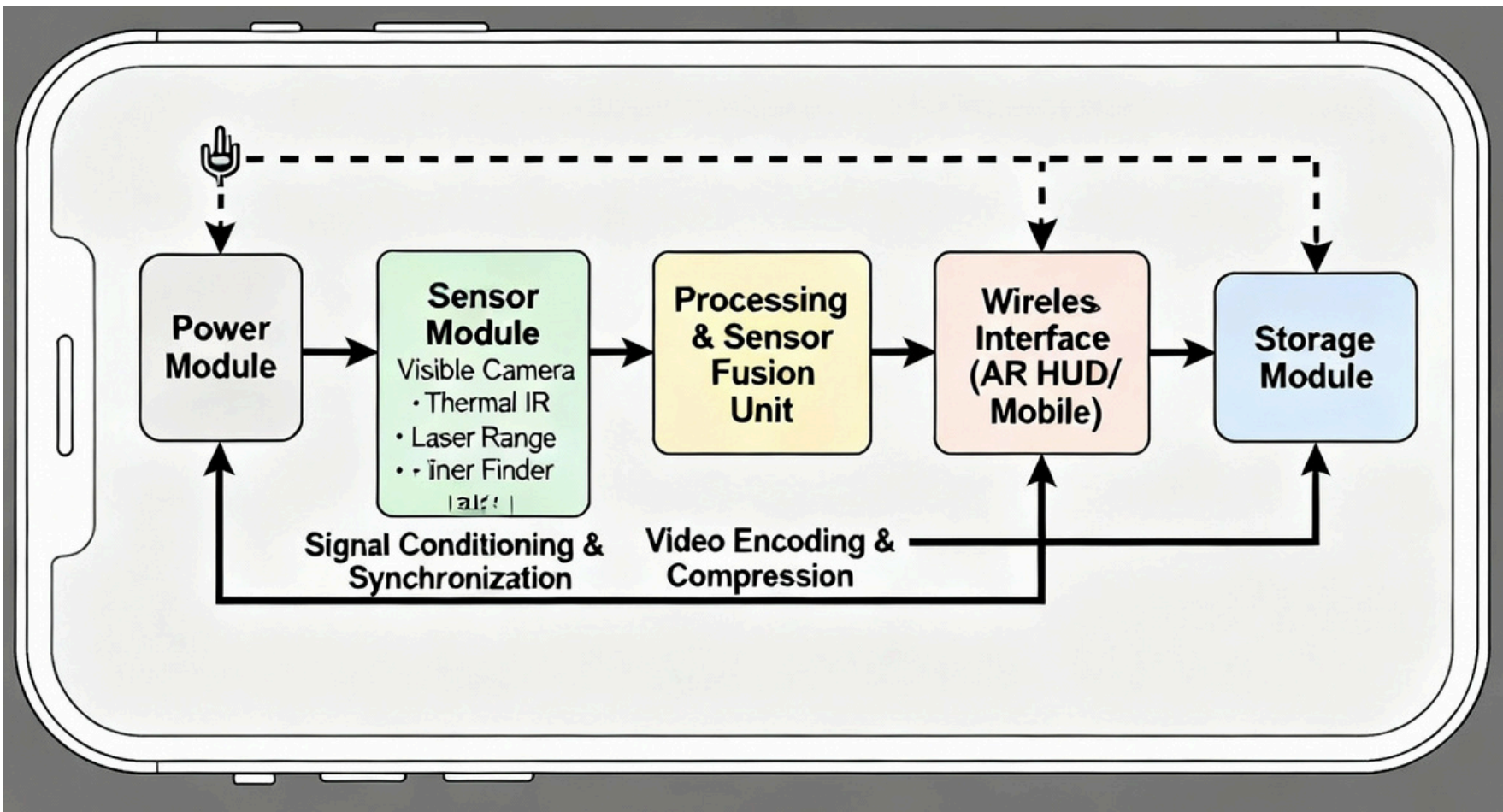
Conceptual Mount strategy



Working Principle of the Surveillance Orb Design



Working And Design Principle of the Surveillance Device- Mobile style



Internal Working

- The battery pack powers all components with regulated energy.
- Sensors capture data: visible light cameras record images, thermal sensors detect heat patterns, laser range finders measure distances, and an inertial measurement unit tracks orientation and movement. Analog sensor signals are converted to digital via ADCs.
- A processor fuses sensor data, performs image processing, and extracts meaningful features. Embedded AI models analyze data for target detection and classification.
- Relevant data is stored locally in memory.
- Wireless modules transmit processed data and raw feeds to ground control via protocols like Wi-Fi or cellular.
- User commands control sensor modes and recording via a connected app or HUD.
- Cooling systems maintain safe operating temperatures.
- IMU data supports device stabilization and orientation, especially on drones or poles.

Components Required

LWIR (Long-Wave Infrared, 8-14 μm):

Well suited for thermal imaging of terrestrial temperature objects and humans in passive sensor mode. It performs best in total darkness and can penetrate fog, haze, and dust but is somewhat limited by atmospheric absorption (especially water vapor).

MWIR (Mid-Wave Infrared, 3-5 μm):

Excels at higher temperature targets with better image contrast and resolution than LWIR. They are good for 1-5 km ranges, especially in low humidity and haze, with better detail and contrast than LWIR.

Visible Light CMOS/CCD Camera

Spectrum: 400–700 nm (visible light)
Purpose: High-resolution daytime imaging and through-window observation
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Near-Infrared (NIR) Camera

Wavelength: 0.7–1.0 μm
Purpose: Low-light imaging with active illumination capability.

Short-Wave Infrared (SWIR) Camera (optional)

Wavelength: 0.9–1.7 μm
Purpose: Better penetration through smoke and some obscurants; material identification.

Laser Range Finder (LRF)

Type: Eye-safe infrared laser
Range: Up to 2 km
Purpose: Measures precise distance and coordinates of detected targets.

Ambient Light Sensor

Purpose: Measures ambient light level for dynamic sensor exposure adjustment.

Environmental Sensors (optional)

Humidity sensor, temperature sensor, and atmospheric pressure sensor
Purpose: To collect environmental data for compensating sensor readings.

Gyroscope and Accelerometer (IMU)

Purpose: Maintains orientation and stabilization data for image stabilization and precise aiming.
Magnetometer
Purpose: Provides compass heading for geolocation and orientation.

Periscope camera

A periscope camera for long-range surveillance uses a folded optical path with prisms or mirrors to achieve a long focal length within a compact form factor. These cameras can provide very high optical zoom—typically up to 100x zoom with lenses up to 1500mm focal length—enabling detailed recognition of targets such as faces and number plates at distances up to 1 km, and observation of activities up to 3 km away.

CMOS Image Sensors

Most widely used in mobile devices.
Integrates photo diode and signal processing on a single chip.
Offers low power consumption, fast readout, and good image quality.
Sizes typically range from 1/4" to 1/1.7" format (approx. 6mm to 9mm diagonal).

Stacked CMOS Sensors

Separate pixel and processor layers stacked vertically.
Enables advanced features like on-chip phase detection autofocus (PDAF) and HDR processing.
Improves image quality while maintaining small sensor size.

Back-Side Illuminated (BSI) Sensors

Sensor circuitry is behind the photodiodes for better light absorption.
Increases sensitivity, especially in low light, making image clearer and brighter.

Sources And Working

- Visible Light Camera
“How Camera Sensors Work – The Basics” by Engineering Explained
<https://www.youtube.com/watch?v=6E4tJ0gOz1k>
- Thermal Infrared Sensor
“How FLIR Thermal Cameras Work” by FLIR Systems
<https://www.youtube.com/watch?v=1t5Zf9jsGrQ>.
- Laser Range Finder (LRF)
“Time of Flight Distance Sensor Tutorial” by GreatScott!
<https://www.youtube.com/watch?v=1j7Z8m3zwpA>
- Inertial Measurement Unit (IMU)
“Understanding IMUs – Accelerometers & Gyroscopes” by DroneBot Workshop
<https://www.youtube.com/watch?v=3bdGNQZ8Qeo>
- GPS/GNSS Receiver
“GPS Fundamentals: How Does GPS Work?” by MIT OpenCourseWare
<https://www.youtube.com/watch?v=RyZxY3C4-Wg>
- Wireless Transmission Module
“LoRa vs. WiFi vs. Bluetooth – Wireless Protocols Explained” by Andreas Spiess
<https://www.youtube.com/watch?v=85IO0BIBkCo>
- Battery Pack and Power Management
“Battery Management System (BMS) Explained” by Walden Kirsch
<https://www.youtube.com/watch?v=q4mQb4--MV4>
- “How FLIR Thermal Cameras Work” – FLIR Systems
<https://www.youtube.com/watch?v=1t5Zf9jsGrQ>
- “MLX90640 32×24 Infrared Thermal Camera Tutorial” – GreatScott!
<https://www.youtube.com/watch?v=C8oO5bXudl8>

Cost Estimation

The total cost of these surveillance devices is calculated by summing the prices of all independent components and modules required to build each unit.

Each major functional part—such as the visible light camera, thermal infrared sensor, IMU, GPS, laser rangefinder, battery pack, wireless transmission module, mounting bracket, enclosure, processor, and optional cooling system—has a distinct market price.

These are individually priced based on their technical specifications and current market rates.

The sum of these individual component costs yields the raw component total for the device.

This calculation does not include bulk discounts, manufacturing overhead, or profit margins—those would be added by manufacturers or integrators in commercial sales.

Component	Description	Price (₹)	Source
Visible Light Camera (high-res)	Main digital imaging sensor (single unit)	7,000 – 22,000	
Thermal Infrared Sensor	Multi-pixel thermal module (256×192 resolution)	6,180 – 28,995	
Laser Range Finder Module	40m–1200m TTL interface module	3,200 – 17,000	
IMU (Accelerometer/Gyro) Module	6-DOF motion sensing	165 – 3,685	
GPS Module	Compact GNSS receiver	950 – 4,500	
Battery Pack	Li-ion, high capacity	4,000 – 9,000	
Wireless TX/RX Module	RF/LoRa/GSM/4G communication hardware	249 – 1,534	
Mounting Bracket	Adjustable drone mounting hardware	69 – 1,250	
Physical Enclosure (IP67/polymer)	Device case, waterproof, rugged	3,000 – 7,000	
AI Processor (edge, optional)	Vision/A.I. processing chip	5,000 – 12,000	
Cooling System (optional)	Heatsink/fan for thermal cameras	500 – 2,000	

Estimated Total Cost by Device Variant

Spherical Omni-Directional Orb

High-end (all advanced modules, edge AI, rugged IP case): ₹48,323 – ₹1,08,464

Mobile Phone-Shaped Device

Typical (high-res visible, thermal, IMU, GNSS, basic wireless, smaller battery): ₹26,544 – ₹75,964

Square/Thin Surveillance Module

Compact (one visible, one thermal, IMU, wireless, basic mounting): ₹19,863 – ₹52,669

The cost of surveillance devices changes over time due to multiple factors, including technology developments, component supply chains, demand cycles, regulations, and global market pressures.

Generally, costs for critical sensors and modules tend to decrease as mass production and innovation improve efficiency—especially for cameras, batteries, wireless modules, and basic processors.

However, rapid adoption of new features (edge AI, advanced analytics, compact sensors) can sometimes create temporary price spikes.